

Empiricist

SEAT EMPTY — still full numbers + smoke honesty · 13+3

CONTRACT

Quote 24/30 · 0.810 I2W · 24.95 Bridge · 3.5× Transfer — refuse to launder DreamGen IF into VLA
SR. Course smoke = frozen 2B shortclip only; do NOT claim PAI 0.768 from smoke.

SPINE: Generative video as physics simulator — Predict2.5 (flow T2W/I2W/V2W) + Reason1 + Transfer2.5 | PARALLEL to latent WMs (Dreamer/JEPA/LeWM)
| synth data · policy eval · closed-loop robotics/AV

Hard numbers table (must-cites)

Paper extract · Ablation Bot folded when present

200M clips	from 35M h; ~4% survive	§2.1
2B / 14B	Predict scales; Transfer 2B	Tab.3 / Abs
93 frames	24 latent @ 16 fps (~5.8 s)	§3.2
PAI T2W 0.768	2B & 14B post Overall	Tab.10
PAI I2W 0.810	2B & 14B post Overall	Tab.11
Robot 24/30	vs baseline 5/30 · base 1/30	Tab.13
Bridge PSNR 24.95	FVD 146; Predict1 21.14/190	Tab.19
TimeEmb best	24.95 > 24.41 > 23.11	Tab.20
AV LET-AP 0.394	vs Transfer1 0.243	Tab.15
Transfer 3.5×	smaller than Transfer1-7B	Abs / §6.1
DreamGen Obj 91.8	IF ≠ VLA SR	Tab.18
Distill 4-step	T2W 0.768→0.764	Tab.7
MFU @4096 H100	2B 36.49% · 14B 33.08%	Tab.9

Caveats Empiricist must voice

Honesty over hype

Tab.18 $\neq$ robot SR

DreamGen scores are video instruction-following (GPT/Qwen judges), not hardware VLA success.

Tab.13 narrow

Single kitchen bimanual task; 3 trials $\times$ 10 scenarios; suggestive n.

RNDS curves

Long-horizon claim primarily Fig.10 curves — no single failure-rate scalar.

GPU-h unpublished

Total pretrain GPU-hours not a single headline; Tab.9 is MFU snapshot on 4096 H100.

Wan Domain edge

Wan2.2-27B-A14B T2W Domain 0.810 vs Cosmos 0.804 — Overall near-tie.

Ablation Bot

File present mid-build $\rightarrow$ fold Tab.20 as only labeled ablation; rest = controlled comparisons.

Course smoke contract (seat empty)

Tiny proxies only — full train = N

RC0	env_smoke clone + import + pins	Y	<0.5 h
RC1	ckpt_probe_shortclip 2B ≥1 clip; NOT PAI 0.768	Y PRIMARY	0.5-4 h
RC2	mode T2W/I2W/V2W three paths; identity check	Partial	1-6 h
RC3	action inject Tab.20 TimeEmb ranking or cite	Partial/cite	0-12 h
RC4	closed_loop_stub finite loss; NOT Tab.13	Partial	1-8 h
RC5	Transfer Edge/Blur infer if ckpt	Partial	1-6 h
—	full WFM / 14B / Tab.13 lab never course	N	weeks+

RC1 PRIMARY — frozen 2B shortclip

Do NOT claim PAI 0.768 from smoke

Pass bar (course)

- Load Predict2.5-2B (prefer post-train); generate 1-4 short clips ($\leq 16-93$ f; may downscale).
- Crash-free; wall-time/clip; VRAM peak logged; mp4 + prompt json saved.
- Optional aesthetic/CLIP proxy non-degenerate (no all-black / NaN).
- NOT required: PAI Overall 0.768 / 0.810 — those need full PAI-Bench.

Fail → adapt (honest)

OOM at 720p → retry 480p/256p / fewer frames / fewer steps / offload. Missing Reason1 → fetch text encoder. Never relabel infer smoke as train ablate.

Tab. 20 — only clean within-family factorial

Ablation Bot A1 / paper labeled Ablation

Method	PSNR↑	SSIM↑	Latent L2↓	FVD↓
TimeEmbedding ★	24.95	0.85	0.28	146
CrossAtten	24.41	0.84	0.28	159
ChannelConcat	23.11	0.78	0.35	267

RC3: prefer released action-cond + cite; lite FT Bridge shard only if inject head swappable — hard cap ~12 GPU-h then stop. Else cite-only.

PAI pre→post + distill (cite)

Train ablations = N for course

0.751→0.768

2B T2W Overall pre→post

0.799→0.810

2B I2W Overall pre→post

0.768→0.764

T2W distill 4-step

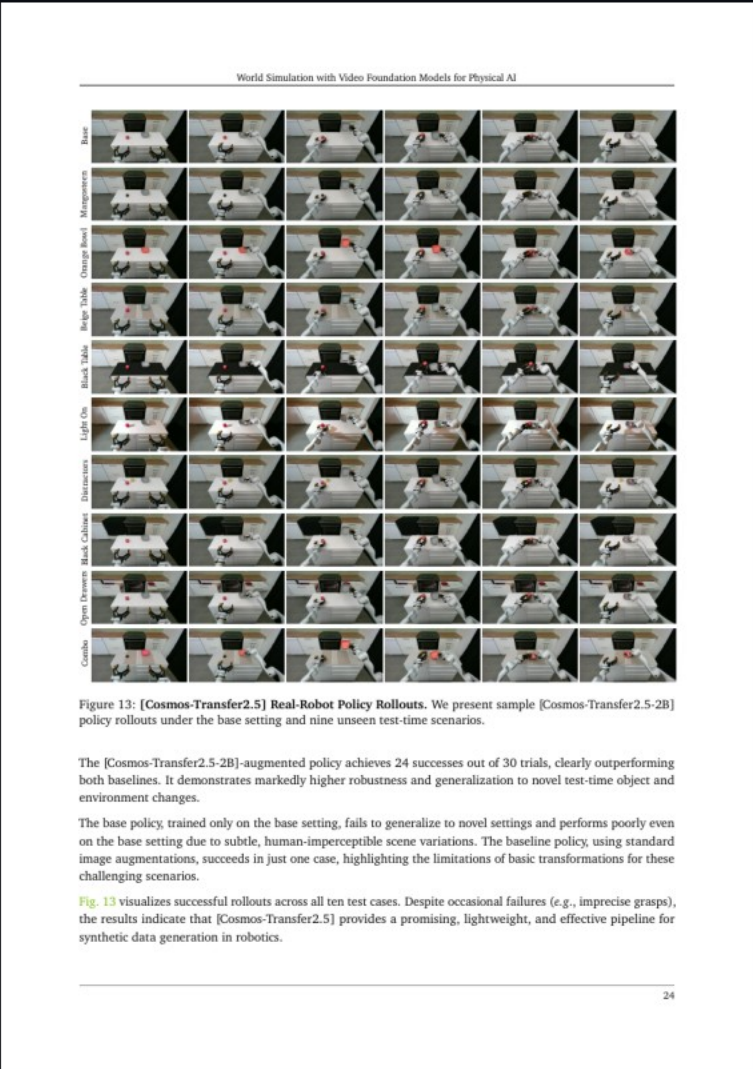
0.810→0.816

I2W distill (student)

Smoke cannot reproduce these without both ckpts + PAI harness. Cite Tabs. 7-8, 10-11.

Fig. 13 — robot rollouts (cite Tab.13)

Empiricist: N lab · Visionary recipe only · numbers stay paper-scoped



24/30 Proposed vs 5/30 baseline vs 1/30 base — single task.

Budget honesty

Course envelope $\leq$ ~8-20 GPU-h infer

RC0-RC1-RC2-RC4	$\leq$ ~8-20 GPU-h infer on 2B	course package
RC3 lite FT	2-12 GPU-h hard cap	then stop / cite
14B any train	N	never
4096 H100 pretrain	N / weeks	Tab.9 MFU class
Tab.13 real robot	N lab	Visionary only
Latent foil LeWM smoke	sibling pack hours	other workspace

Follow-up experiments (hypothesis-ready)

Not launched in quiet run

F1 Matched wall-clock CEM: LeWM/2-AC latent vs distilled 4-step Cosmos pixel rollout cost.

F2 Expand Tab.13 to multi-task / BEHAVIOR-style language goals.

F3 Train VLA on DreamGen data → report REAL SR, not only video IF.

F4 Physics probes (VideoPhy / IntPhys-style) on Predict2.5 vs Wan.

F5 Transfer RNDS on 5–10 min horizons for AV closed-loop stress.

F6 If Ablation Bot lands mid-build: re-rank RC3/A1 priority (already #1).

Foil note vs LeWM / JEPA (conceptual)

Not a paper bake-off — Archaeologist framing

LATENT (sibling packs)

LeWM/I-JEPA trainable on 1 GPU; CEM ~0.98s
moral; decoder-off can help planning.

Course can actually smoke train knobs.

GENERATIVE (this pack)

DiT video roll $\gg$ latent cost; reserved for synth
data / photoreal eval.

Course = infer-only probes · full train N.

H-foil (#12)

On tiny control proxy, LeWM/JEPA latent plan is orders of magnitude cheaper than DiT video roll — generative
WFM for synth/eval, not inner MPC. Do NOT claim paper comparison.

Numbers tattoo — pin these (KanBot must-cites)

Shared spine across all five seats · paper numbers only

24/30

Real2Real robot vs 5/30

0.768

PAI T2W Overall post

0.810

PAI I2W Overall post

24.95

Bridge AC PSNR

IF $\neq$ SR

DreamGen Tab.18 $\neq$ VLA

0.394

AV LET-AP Transfer2.5

3.5×

Transfer smaller vs T1

200M

curated video clips

2B / 14B

Predict scales

93f @16fps

~5.8 s generation

PARALLEL

not JEPA child

GENERATIVE

pixel WFM pole

Empiricist — close

Takeaways + exam bite

1. Pin must-cites: 24/30 · 0.768/0.810 · 24.95 · 0.394 · 3.5× · 200M · 2B/14B · 93f · distill 4-step · MFU.
2. Seat empty → smoke = RC1 frozen 2B shortclip; never claim PAI from smoke.
3. Refuse DreamGen IF→VLA SR laundering; Tab.13 stays single-task scoped.
4. Follow-ups ready: matched wall-clock vs LeWM; DreamGen→real SR; physics probes.

EXAM BITE

Name three must-cite numbers and one smoke rule (RC1 does not prove PAI 0.768).